6. Write a Servlet to demonstrate session tracking using HttpSession. Implement a simple login system where the user's session is tracked.

AIM

To create a web application that demonstrates session tracking using the `HttpSession` interface. The application will maintain a user's login state across multiple pages of the Weather Information portal.

ALGORITHM

1. **Input:** The user enters their credentials on the login page.
2. **Session Creation:**
   - Upon successful authentication, the Servlet creates a session using `request.getSession()`.
   - The user's identity is stored as a session attribute (e.g., `session.setAttribute("user", name)`).
3. **Session Tracking:**
   - When the user navigates to the Weather Dashboard, the Servlet checks if a valid session exists.
   - If `session.getAttribute("user")` is not null, the dashboard is displayed; otherwise, the user is redirected to the login page.
4. **Session Termination:** A "Logout" button is provided to invalidate the session using `session.invalidate()`, ensuring the user is logged out.
5. **Deployment:** Compile the Servlet, map it in the `web.xml` or via annotations, and deploy it to Apache Tomcat.

SOURCE CODE

A. HTML File (index.html)

This file provides the login interface for the weather portal.

HTML

```
<!DOCTYPE html>
<html>
<head>
    <title>Weather Portal | Login</title>
    <style>
        body { font-family: 'Segoe UI', sans-serif; background: #f0f2f5; display: flex;
justify-content: center; align-items: center; height: 100vh; margin: 0; }
        .login-card { background: white; padding: 30px; border-radius: 12px; box-
shadow: 0 4px 12px rgba(0,0,0,0.1); width: 300px; text-align: center; }
        input { width: 100%; padding: 10px; margin: 10px 0; border: 1px solid #ddd;
border-radius: 6px; box-sizing: border-box; }
        .btn { width: 100%; padding: 10px; background: #007bff; color: white; border:
none; border-radius: 6px; cursor: pointer; }
    </style>
</head>
<body>
    <div class="login-card">
        <h2>🌤 Weather Login</h2>
        <form action="SessionServlet" method="POST">
            <input type="text" name="name" placeholder="Username" required>
            <input type="password" name="password" placeholder="Password" required>
```

```
            <input type="submit" class="btn" value="Login">
        </form>
    </div>
</body>
</html>
```

Java File (SessionServlet.java)

This Servlet manages the session creation and the weather dashboard.

Java

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.WebServlet;

@WebServlet("/SessionServlet")
public class SessionServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String pass = request.getParameter("password");

        if(pass.equals("admin123")) {
            // Create a session and store user info
            HttpSession session = request.getSession();
            session.setAttribute("user", name);
            response.sendRedirect("DashboardServlet");
        } else {
            out.print("<p style='color:red'>Invalid Password!</p>");
            request.getRequestDispatcher("index.html").include(request, response);
        }
    }
}
```

Java File (DashboardServlet.java)

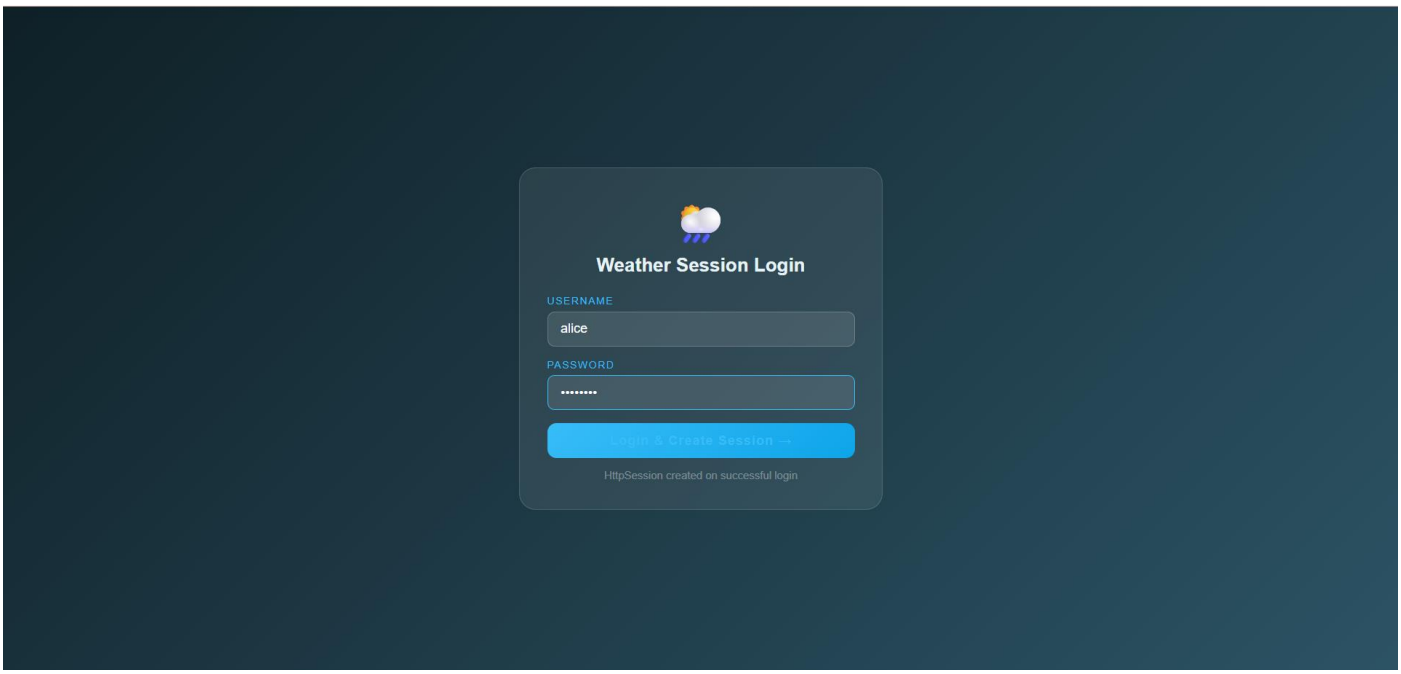
This Servlet verifies the session before showing weather data.

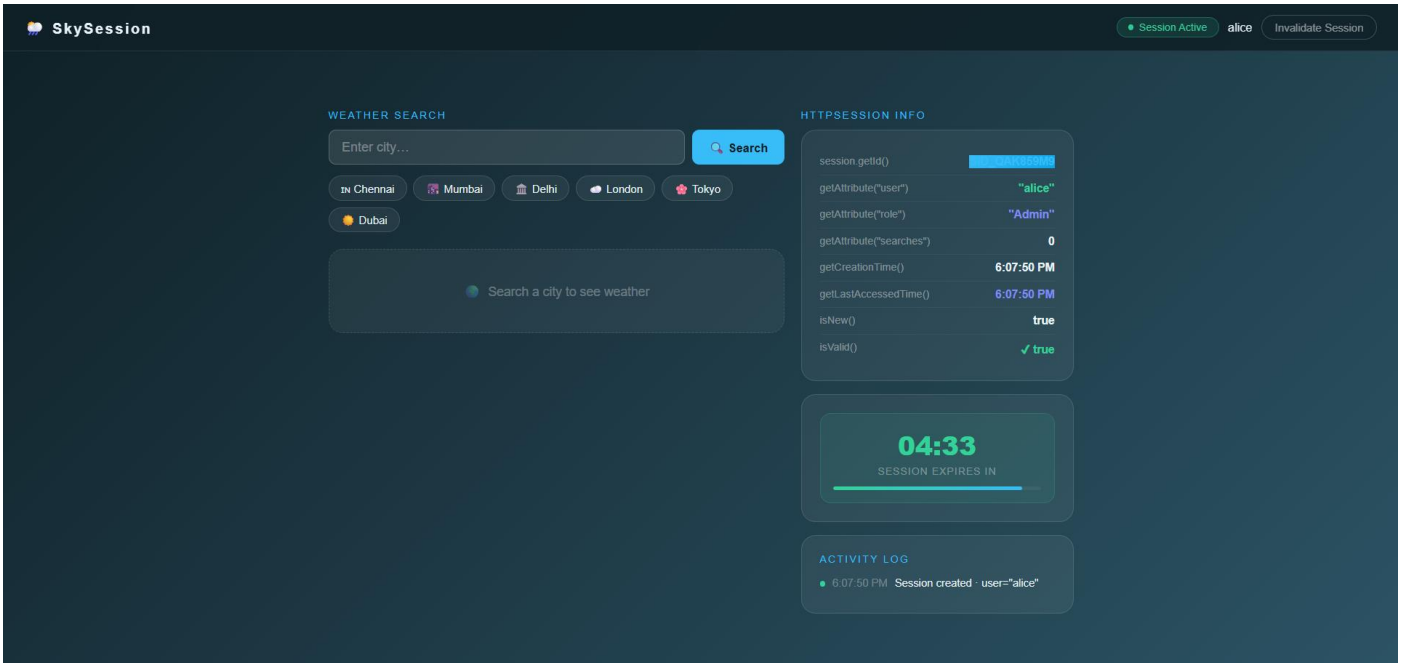
Java

```
@WebServlet("/DashboardServlet")
public class DashboardServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        HttpSession session = request.getSession(false);
        if (session != null && session.getAttribute("user") != null) {
            String name = (String) session.getAttribute("user");
            out.println("<body style='text-align:center; padding-top:50px;'>");
            out.println("<h1>Welcome to Weather Dashboard, " + name + "!</h1>");
            out.println("<p>Current Weather: 25°C Sunny 🌞</p>");
            out.println("<a href='LogoutServlet'>Logout</a>");
            out.println("</body>");
        } else {
            out.print("Please login first!");
            request.getRequestDispatcher("index.html").include(request, response);
        }
    }
}
```

OUTPUT:


Weather Session Login
USERNAME
alice
PASSWORD
Login & Create Session →
HttpSession created on successful login


SkySession
Session Active
alice
Invalidate Session
WEATHER SEARCH
Enter city...
Search
IN Chennai
Mumbai
Delhi
London
Tokyo
Dubai
Search a city to see weather
HTTPSESSION INFO
session.getId()
getAttribute("user")
"alice"
getAttribute("role")
"Admin"
getAttribute("searches")
0
getCreationTime()
6:07:50 PM
getLastAccessedTime()
6:07:50 PM
isNew()
true
isValid()
✓ true
04:33
SESSION EXPIRES IN
ACTIVITY LOG
6:07:50 PM Session created · user="alice"

RESULT

A Java Servlet application was successfully implemented to demonstrate session tracking using `HttpSession`. The application effectively maintained user data across different web pages within the Weather Information portal.